TRUNCATION ISOMORPHISMS OFF AN OPEN DETECTED BY ORTHOGONALITY TO THE KOSZUL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-affine-compare-bounded is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Triangle splitting off the open. structural (distinguished triangle of a restriction). N -> E -> Rj_*(E|_U) -> N[1] with N supported on Z = X - U.
- Multiplication triangles of Koszul complexes. More on Algebra, lemma koszul-mult (cross-chapter, named). Relates Hom against K to Hom against the powers K_e.
- Representing classes by dual Koszul complexes. lemma-represent-cohomology-class-on-closed (carded L26768). For N supported on Z, Hom(K_e[-n], N) = 0 for all e iff H^n(X, N) = 0 in the relevant range.
- Affine comparison of D_QCoh with module complexes. lemma-affine-compare-bounded (same chapter, NOT carded, declared external). N is determined by the complex R Gamma(X, N) of A-modules.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E3: sources/stacks-perfect/pinned-original-source.tex, source lines 2374–2374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 062C: focused-seeds/geometry/S9-perf2-orthogonal-koszul-first-variant/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08CZ: sources/stacks-perfect/pinned-original-source.tex, source lines 2116–2116. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 08CZ denote $j : U \to X$ the open immersion and let K be the perfect object of $D(\mathcal{O}_X)$ corresponding to the Koszul complex on $f_1, \ldots, f_r$ over A. Let $E \in D_{QCoh}(\mathcal{O}_X)$ and $a \in \mathbf{Z}$. Consider the following conditions*

1. *The canonical map $\tau_{\geq a}E \to \tau_{\geq a}Rj_*(E|_U)$ is an isomorphism.*
2. *We have $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], E) = 0$ for all $n \geq a$.*

Then (2) implies (1) and (1) implies (2) with a replaced by $a + 1$.

Proof. Choose a distinguished triangle $N \to E \to Rj_*(E|_U) \to N[1]$. Then (1) implies $\tau_{\geq a+1}N = 0$ and (1) is implied by $\tau_{\geq a}N = 0$. Observe that

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], Rj_*(E|_U)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(K|_U[-n], E) = 0$$

for all n as $K|_U = 0$. Thus (2) is equivalent to $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], N) = 0$ for all $n \geq a$. Observe that there are distinguished triangles

$$K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'+e_i''}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i''}, \ldots, f_r^{e_r}) \to \ldots$$

of Koszul complexes, see More on Algebra, Lemma 062C. Hence $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], N) = 0$ for all $n \geq a$ is equivalent to $\mathrm{Hom}_{D(\mathcal{O}_X)}(K_e[-n], N) = 0$ for all $n \geq a$ and all $e \geq 1$ with K_e as in Lemma 08E3. Since $N|_U = 0$, that lemma implies that this in turn is equivalent to $H^n(X, N) = 0$ for $n \geq a$. We conclude that (2) is equivalent to $\tau_{\geq a}N = 0$ since N is determined by the complex of A-modules $R\Gamma(X, N)$, see Lemma 06Z0. Thus we find that our lemma is true. □